# Saurabh Surashe

MTS III at Mavenir Systems

*+91-7276561155* || surashesaurabh21@gmail.com || *LinkedIn*

## Professional Summary

C/C++ Systems Software Engineer with **4+ years of experience** developing high-performance, latency-sensitive software for carrier-grade communication and networking systems. Experienced in **Modern C++, Linux systems programming, concurrent programming, inter-process communication, protocol processing, performance optimization, and production debugging**. Designed and delivered **130+ production features, protocol enhancements, performance optimizations, and reliability fixes** across large-scale commercial software releases.

Strong background in Layer-2 scheduling, networked systems, real-time software, and distributed software components, with a focus on **networking infrastructure, systems software, high-performance C++, and connectivity platforms**.

## Core Competencies

**Programming:** C, C++17, STL, Python, Java
**Linux & Systems:** Linux, pthreads, Multithreading, Shared Memory IPC, Process/Thread Synchronization, GDB, Linux perf, Intel VTune
**Networking:** TCP/IP, UDP, Networked Systems, Protocol Processing, Wireshark, Packet Analysis
**Wireless Networking:** NB-IoT, LTE, Non-Terrestrial Networks (NTN), MAC, RRC, S1AP, ASN.1
**Software Engineering:** Concurrent Programming, Distributed Systems, Event-Driven Programming, Real-Time Systems, Low-Latency Systems, Performance Optimization, Root-Cause Analysis
**AI / Machine Learning:** Sentence Transformers, Semantic Search, Embeddings, TensorFlow, Keras

## Professional Experience

**Mavenir Systems Pvt. Ltd.** July 2022 – Present
*Member of Technical Staff I → Member of Technical Staff III* *Bengaluru, India*

- Designed and developed high-performance **C/C++ systems software on Linux** for carrier-grade communication and networking platforms, contributing to scheduler architecture, protocol processing, concurrency, performance optimization, debugging, and production software delivery.
- Redesigned a latency-sensitive **MAC scheduler resource-allocation workflow**, improving uplink throughput by approximately **40%** while maintaining deterministic scheduling deadlines for real-time workloads.
- Delivered more than **130 production software features, protocol enhancements, performance optimizations, and critical customer fixes** across multiple commercial software releases.
- Architected Layer-2 scheduling enhancements for **3GPP Release-17 Non-Terrestrial Networks (NTN)**, implementing Timing Advance computation, dynamic satellite ephemeris management, and interlaced scheduling for GEO and LEO satellite communication.
- Designed configurable **link adaptation algorithms** using radio-quality measurements and HARQ feedback to optimize throughput and communication reliability under varying channel conditions.
- Designed a low-latency **shared-memory IPC** framework enabling runtime propagation of system state and satellite ephemeris information across distributed software components without service interruption.
- Developed concurrent software using **pthreads, synchronization mechanisms, shared memory, and event-driven processing** for performance-sensitive real-time workloads.
- Implemented more than **60 production KPIs** providing visibility into scheduler behavior, protocol execution, resource utilization, system performance, and software health.
- Profiled and optimized latency-sensitive execution paths using **Intel VTune and GDB**, identifying CPU bottlenecks and improving runtime efficiency.
- Performed systematic root-cause analysis of complex production issues using **runtime logs, protocol traces, Wireshark, memory analysis, execution profiling, and low-level debugging**, resolving performance, reliability, synchronization, and protocol-related issues.
- Developed protocol-processing software across **MAC, RRC, S1AP, and ASN.1** layers and integrated software components across multiple system modules and network interfaces.
- Collaborated with Layer-1, Platform, QA, DevOps, and Customer Engineering teams throughout architecture, implementation, validation, deployment, and production support.
- Mentored junior engineers through technical design discussions, debugging sessions, code reviews, and knowledge sharing.

## Projects

**AI-Powered Job Discovery & Matching Platform** | *Python, FastAPI, SQL, Sentence Transformers, NumPy*

- Developed a modular Python platform automating job discovery and semantic resume-to-job matching using the `all-MiniLM-L6-v2` transformer model and embedding-based similarity scoring.
- Built asynchronous scraping pipelines using `asyncio` to concurrently fetch and process job opportunities from Workday, Greenhouse, Eightfold, and SmartRecruiters ATS platforms.
- Implemented vector-based semantic similarity search using NumPy for efficient resume-to-job matching without database-side vector extensions.
- Designed REST APIs using FastAPI for resume parsing, job indexing, matching configuration, and automated job processing.
- Built automated notification workflows delivering HTML job digests through SMTP and scheduled serverless processing pipelines.

## Achievements

**Mavenir Spotlight Award** – Received multiple times for high-impact production features, technical contributions, and successful customer software releases.

Co-authored **"Enhancing Cyber Security: Malicious Web Page Detection using Machine and Deep Learning"**, AINA 2025 (*SpringerLink*)

Co-authored **"Privacy and Security Comparison of Web Browsers: A Review"**, AINA 2022 (*SpringerLink*)

## Education

**National Institute of Technology Surathkal** — Karnataka
*Master of Technology in Computational Data Science* — *June 2020 – May 2022*

**MIT College of Engineering Pune** — Maharashtra
*Bachelor of Engineering in Mechanical Engineering* — *June 2015 – May 2019*